

A 1-mW Class-AB Amplifier With -101 dB THD+N for High-Fidelity $16\ \Omega$ Headphones in 65-nm CMOS

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Abstract—A low-power class-AB amplifier for high-fidelity headphones is presented. The linearity of a class-AB amplifier strongly depends on the loop gain over the audio band and voltage swing. In this paper, a cascode-driven mesh technique limits the number of high-swing nodes in the output stage, thereby preserving the loop gain of the preceding gain stages. A frequency compensation scheme is proposed that extends the unity-gain bandwidth (UGB) and increases the loop gain at 20 kHz. Combining these two techniques, the amplifier achieves -101.4 -dB total harmonic distortion plus noise (THD+N) over the full audio band, the lowest ever reported nonlinearity among sub-milliwatt CMOS class-AB amplifiers. The amplifier delivers 51.2-mW peak power to a $16\ \Omega \parallel 0.33$ -nF load while consuming 0.97 mW of total static power. It has UGB of 12.3 MHz and SNR of 108 dB. These results are consistent when measured over eight test chips. Compared to the recent prior art, the amplifier exhibits >12 dB better linearity and >14 -dB dynamic range. This results in $2.5\times$ improvement in figure of merit (FOM = Peak Load Power/(Quiescent Power $\times$ THD+N[%])). It occupies $0.16\ \text{mm}^2$ in a standard 65-nm CMOS.

Index Terms—Audio amplifier, cascode-driven mesh bias, class-AB, frequency compensation, headphone driver, high-fidelity (hi-fi), high-linearity, low-power, standard digital CMOS.

I. INTRODUCTION

THE ability to support high-fidelity (hi-fi) audio has emerged as an indispensable feature of modern mobile devices such as smartphones and portable media players [1]. Driven by a surge in on-demand music streaming and manufacturers' continual efforts to distinguish themselves, today's mobile devices are adapting to support high-resolution audio standards. The audio quality, commonly measured in terms of total harmonic distortion plus noise (THD+N), should be better than -100 dB for hi-fi audio playback. As mobile devices are battery operated, the favored solutions to improve THD+N are the ones that also minimize power consumption.

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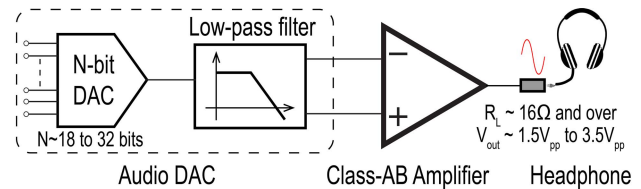
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Performance of each block in the signal path

Technology	Audio DAC			Class-AB amplifier			Audio signal path	
	Ref.	THD [dB]	Power [mW]	Ref.	THD [dB]	Power [mW]	THD [dB]	Power [mW]
Discrete	[5]	-115	26	[6]	-106	19	-103	45
Integrated	[7]	-103	0.7	[8]	-89	0.4	-86	1.1

Fig. 1. Typical hi-fi audio signal path and its performance metrics.¹

To identify the dominant source of distortion and power consumption, Fig. 1 shows a typical hi-fi audio signal path used in mobile devices. It comprises of a high-resolution audio digital-to-analog converter (DAC), a low-pass filter, and a headphone driver [2], [3]. The class-AB amplifiers are preferred for headphone drivers due to ease of integration, superior audio quality, better power supply noise rejection, and no electromagnetic interference (EMI) issues prevalent in digital switching amplifiers like class-D [4]. An audio path realized assuming a cascade connection of a discrete hi-fi DAC [5] and discrete hi-fi class-AB amplifier [6] has linearity sufficient for hi-fi audio. Unfortunately, it consumes significant power, which shortens the battery life and limits the music playback time. Consequently, such hi-fi audio paths are not integrable and remain as an external pluggable module that augments the mobile devices. Alternatively, an audio path implementation with a CMOS DAC [7] and a CMOS class-AB amplifier [8] integrated on a single chip, has significantly lower power consumption. However, it achieves only moderate linearity (<90 dB) limited by the THD+N of the class-AB amplifier. Clearly, to integrate the hi-fi audio path in mobile devices, improving the linearity of a CMOS Class-AB amplifier while consuming low power is necessary [9]. To this end, the target THD+N for the hi-fi class-AB amplifier is <-100 dB over the full audio band (20 Hz–20 kHz) while keeping power consumption below 2 mW.

¹ Assumptions to simplify calculation: Only HD2 and HD3 of equal magnitude are considered. DAC and the amplifier both have $1.0\text{-}V_{\text{rms}}$ output.

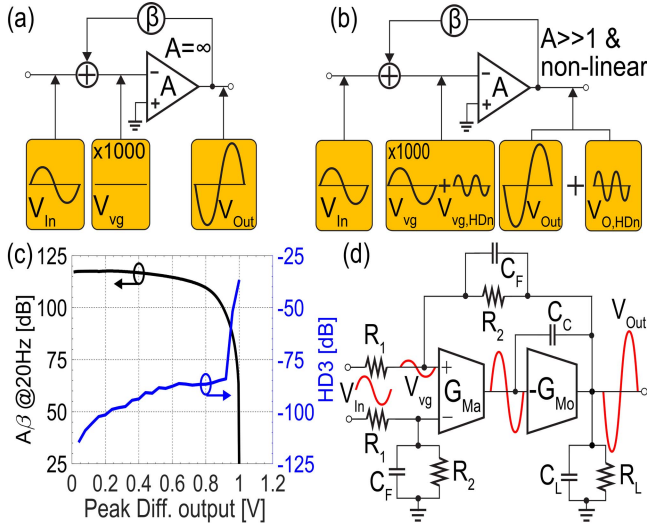


Fig. 2. (a) Ideal and (b) non-ideal amplifier under negative feedback [13], (c) variation in loop gain and $HD3$ with output voltage swing for $\beta = 0.25$, and (d) typical two-stage realization of a class-AB amplifier.

The distortion in a class-AB amplifier is dominated by the output stage [10]. This paper introduces two new techniques that suppress distortion by conserving and boosting the loop gain. A low-power class-AB output stage requires a high-swing gate drive. Gain stages that generate such a high-swing drive suffer a steep loss of loop gain (covered in Section II). To address this issue, a cascode-driven mesh biasing technique is proposed whose design details and tradeoffs are discussed in Section III. Furthermore, Section IV describes a new *damping factor controlled multipath hybrid nested Miller compensation (DFC-MHNMC)* scheme that increases the loop gain at 20 kHz by extending the unity-gain bandwidth (UGB). It is realized by applying a damping network that supports large voltage swing [11] to a conventional MHNMC scheme [12]. Section V details the circuit implementation of the amplifier. Measured results discussed in Section VI evaluate the effectiveness of these two proposed solutions. Section VII provides concluding remarks.

II. DESIGN CHALLENGES

An ideal amplifier under negative feedback shown in Fig. 2(a) is typically realized with a linear feedback factor β and an amplifier with infinite open-loop gain A . Here, the output is simply related to the input by $V_{out} = -V_{in}/\beta$ and no distortion is generated. However, in a real-life amplifier, the gain, A , is finite and non-linear as it varies with frequency and voltage swings. Such an amplifier generates distortion at the output ($V_{O,HDn}$), which is forced into the virtual ground node by the feedback as shown in Fig. 2(b). The distortion at these two nodes is related as: $V_{vg,HDn} = \beta V_{O,HDn}$ [13]. Using this premise, the second-order ($HD2$) and third-order ($HD3$) harmonic distortion under feedback is related to the open-loop distortion as [14]

$$HD2 = \frac{HD2_{open}}{A\beta} \quad \text{and} \quad HD3 = \left| \frac{HD3_{open} - 2HD2_{open}^2}{A\beta} \right| \approx \frac{HD3_{open}}{A\beta}. \quad (1)$$

For the symmetric differential amplifiers, $HD2$ is much lower than $HD3$. Hence, assuming $HD3_{open} \gg 2(HD2_{open})^2$ in (1) is valid. The essence of (1) is that increasing the loop gain lowers harmonic distortion. Unfortunately, the loop gain is a strong function of frequency [15] and voltage swing. To illustrate this effect, a four-stage MHNMC stabilized amplifier with a conventional class-AB output stage is simulated with a β of 0.25 and a supply of ± 1.1 V [12], [16]. Its loop gain and $HD3$ as a function of output voltage swing are shown in Fig. 2(c). As the output swings from 20 mV_{pp} to 2V_{pp}, the loop gain drops from a 120-dB level down to a 20 dB. This degrades $HD3$ because the devices tied to the high-swing nodes have larger coefficients of nonlinearity as they experience increased variation in transconductance (g_m) and output impedance (g_{ds}). If the loop gain is somehow held a constant, this linearity degradation can be avoided.

Recent works have tried to address this issue. An enhanced replica bias circuit in [2] prevents bias variation, and thus, loop-gain variation due to process and device mismatch. It robustly biases the class-AB stage at low-quiescent current, which is set to achieve a target THD. However, it still cannot stop loop-gain variation with voltage swing or frequency. As a result, it achieves only typical THD+N (<80 dB). A modified version in [3] realizes a class-AB amplifier with a popular two-stage topology similar to Fig. 2(d). Here, the output stage, G_{Mo} , is designed to drive a large voltage swing while operating with low power. As the gain of G_{Mo} is around unity, the preceding gain stage G_{Ma} , also experiences sizeable voltage swing making it lose its loop gain [Fig. 2(c)]. In addition, the loop gain of two-stage topology is insufficient, limiting the linearity to moderate (<90 dB) level. In [8] and [11], a three-stage topology is used to increase the loop gain. However, the techniques used to save power, such as low-quiescent current and multi-voltage domain, create a number of high-swing nodes resulting in moderate distortion.

This paper shows a four-stage amplifier with cascode-driven mesh biasing that preserves the loop gain by isolating the gain stages from the high-swing nodes in the output stage. The proposed DFC-MHNMC compensation extends UGB by using damping factor control (DFC) to reduce the level of Miller nesting. In addition, wherever possible, active compensation is used instead of Miller narrowbanding to avoid bandwidth reduction. To reduce power, a multi-voltage domain solution, as in [11] and [17], is used. However, this paper uses apt design choices to avoid unnecessary circuits, such as level shifters required to cross voltage domains. This not only saves power but also avoids additional sources of distortion.

III. CASCODE-DRIVEN CLASS-AB MESH BIASING

This section examines the use of a cascode-driven mesh to address the nonlinear distortion issues mentioned above.

A. Conventional Class-AB Mesh

A mesh circuit formed by two head-to-tail connected transistors, M_1 and M_2 in Fig. 3(a), bias the output stage in the class-AB mode [18]. It is the simplest solution to the level-shifting problem in the output stages. The devices M_{D1} and

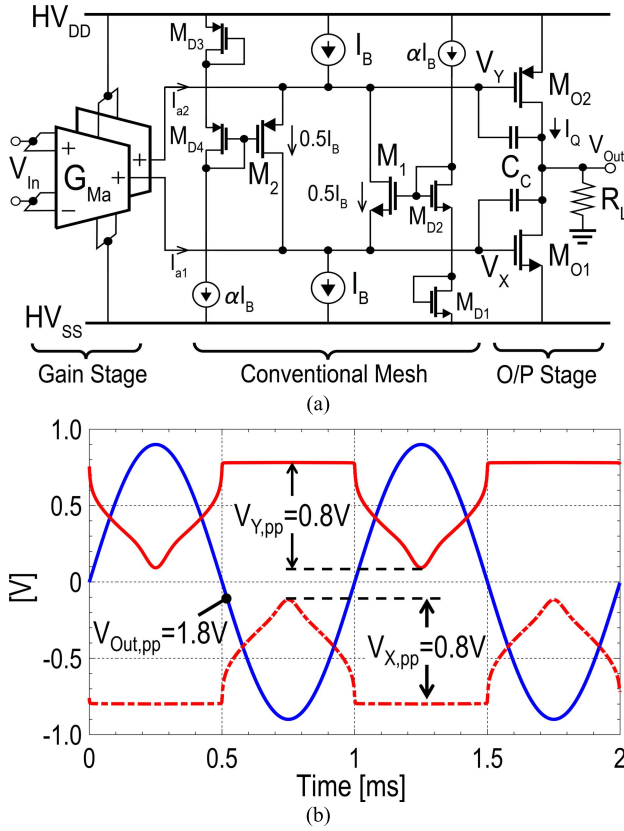


Fig. 3. Operation of conventional class-AB mesh biasing circuit [18]. (a) Circuit diagram of a conventional mesh. (b) Transient plot of $V_{X,Y}$ for 1.8-V peak-to-peak V_{Out} swing.

M_{D2} (M_{D3} , M_{D4}), and M_1 and M_{O1} (M_2 , M_{O2}) form a quadratic translinear loop [19]. Hence, their gate–source voltages are equal, i.e., $V_{gs,MD1} + V_{gs,MD2} = V_{gs,M1} + V_{gs,MO1}$. The output quiescent current, I_Q , is set by the ratio of widths of M_{O1} over M_{D1} . The current sources I_B bias the mesh devices, which are sized such that $I_{M1} = I_{M2} = 0.5I_B$, at the quiescent point. To save power, the diodes M_{D1} – M_{D4} are biased at α times lower current than the mesh devices. As shown in Fig. 3(b), depending on the input signal, the gain stage G_{Ma} sources or sinks current to drive the nodes X and Y . A sinking current steers I_B into M_1 to cut off M_2 creating considerable swing in V_Y to pull-up the output. Similarly, when G_{Ma} sources current, V_X increases to pull-down the output while I_B is steered to M_2 , cutting M_1 off. During these excursions, the output devices M_{O1} and M_{O2} are never turned off and return to their nominal bias at the end.

For smaller load values ($R_L = 16 \Omega$), significant output swing is required to deliver reasonable power. This increases the required gate swing, $V_{X,Y}$, as it depends upon the difference between I_Q and the required peak load. To save power, I_Q is kept low, reducing the gain of the output stage close to unity. This further increases the required swing on $V_{X,Y}$. In addition, as evident from Fig. 3(b), V_X and V_Y are asymmetric and swing from HV_{SS} and HV_{DD} power rails, respectively. Consequently, the gain stages connected to such nodes with asymmetric high-swing requirements suffer a steep loss of loop gain. Finally, the Miller capacitor C_C is terminated to

$V_{X,Y}$ as a virtual ground. However, the presence of large swing generates diode nonlinearity that appears in series with the output. This creates an additional source of distortion [16].

B. Proposed Cascode-Driven Class-AB Mesh

In the proposed cascode-driven mesh shown in Fig. 4(a), the gain stage, G_{Mb} , is isolated from high-swing nodes X and Y using the shielding property of cascodes. As shown in Fig. 4(a), the mesh of M_1 and M_2 is driven by cascodes M_3 and M_4 . The translinear relationship is still kept intact, which sets I_Q as before. The gain stage G_{Mb} now drives $V_{P,Q}$ instead of $V_{X,Y}$ [Fig. 4(b)]. As $V_{P,Q}$ swing is much lower, it is easier to support it. Thus, unlike G_{Ma} , G_{Mb} does not experience steep loss in the loop gain, thereby improving its distortion performance. Furthermore, as the output swing requirement of G_{Mb} is reduced, it can be powered from lower power rails (AV_{DD} and AV_{SS}) reducing the total power. In addition, unlike G_{Ma} , the required swing at nodes P and Q is symmetric around the ground. This enables voltage domain crossing from AV_{DD} and AV_{SS} to HV_{DD} and HV_{SS} without requiring level shifters, as used in [11] and [17]. This further saves power and avoids unnecessary sources of distortion. Terminating the Miller capacitor C_C to $V_{P,Q}$ avoids the diode distortion and also achieves higher gain bandwidth than the traditional Miller compensation due to active compensation provided by M_3 and M_4 [20]. Finally, the large potential difference between HV_{DD} and HV_{SS} can cause gate–drain oxide breakdown. Cascodes M_3 and M_4 alleviate this issue by sharing the drain–source voltage.

Fig. 4(b) shows the transient operation of cascode-driven mesh. Here, the current entering a node is considered positive while that leaving a node is negative. The minimum selector devices, M_{S1} and M_{S2} , in Fig. 4(a) are used to avoid large voltage swings on $V_{P,Q}$ due to a large input signal. The gain stage G_{Mb} pulls the output high by sinking current I_{b2} from node Q . This drops V_Q and shuts M_2 and M_4 off steering all of the bias current I_B through M_1 and M_3 . As V_Q continues to drop, M_{S1} weakly turns on to limit the swing. Similarly, to pull the output low G_{Mb} sources I_{b1} , which pulls V_P high and steers I_B to M_2 and M_4 . This turns M_1 and M_3 off while weakly turning on M_{S2} . The current sourced and sunk by G_{Mb} is symmetric and equal, which ensures V_P and V_Q have an opposite but equal swing. At the quiescent point, M_{S1} and M_{S2} consume no static current as their gate–source voltage is set to zero by their respective bias voltages V_{b3} and V_{b4} . They are activated only when V_P (V_Q) crosses the threshold of M_{S1} (M_{S2}). Furthermore, these devices are sized to deliver a maximum of $15 \mu A$ when fully turned on, which is a fraction of I_Q ($\sim 5.5\%$). When $V_{P,Q}$ return to its nominal quiescent value, the current selectors are shut-off again.

C. Signal Analysis of Cascode-Driven Mesh

Fig. 5 shows the equivalent circuit of conventional- and cascode-driven mesh. Here, g_{mfloat} represents g_m of the mesh devices M_1 and M_2 , g_{mCasc} is the g_m of the cascodes M_3 and M_4 , and Z_T is the output impedance of the mesh bias

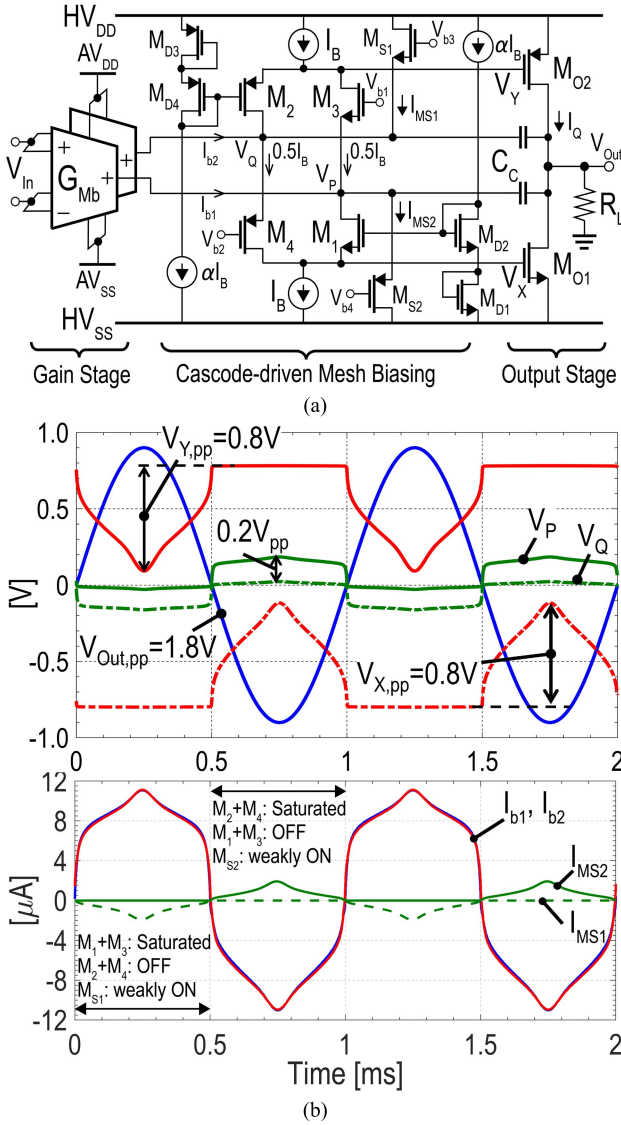


Fig. 4. Operation of cascode-driven class-AB mesh biasing circuit. (a) Circuit diagram of proposed cascode-driven class-AB mesh. (b) Transient plot of key voltage and current signals for 1.8 V $V_{Out,pp}$.

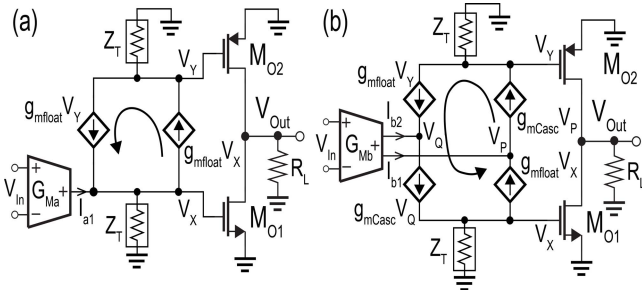


Fig. 5. Equivalent circuit of (a) conventional and (b) cascode-driven mesh.

current source, I_B . The input impedance of the conventional mesh circuit is found to be

$$Z_{in,conv} = \frac{V_X}{I_{a1}} \approx Z_T. \quad (2)$$

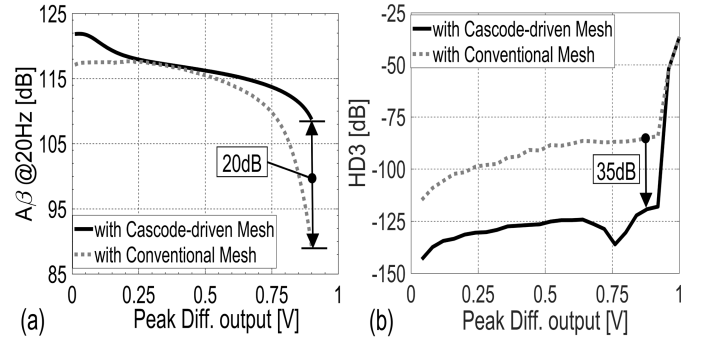


Fig. 6. Simulated (a) $A\beta$ and (b) HD3 with differential output swing at 20 Hz.

The mesh devices circulate the signal current from one branch to other without altering it. As a result, the output impedance of M_1 and M_2 (r_{O1} and r_{O2}) does not affect (2). Using (2) and $g_{mfloat}(r_{O1}||r_{O2}) \gg 1$, it is evident that the input impedance of the conventional mesh is limited by the output impedance of the current source I_B and not by the mesh devices. The open-loop voltage gain from V_{in} to V_{out} is

$$G_{conv} = \frac{V_{out}}{V_{in}} = (G_{Ma}Z_T)(G_{Mo}R_L) \quad (3)$$

where $G_{Mo} (= g_{m,O1} + g_{m,O2})$ is the g_m of the output stage.

Furthermore, in the cascode-driven mesh shown in Fig. 5(b), it can be readily seen that V_X and V_Y , and V_P and V_Q are a copy of each other. The relation between $V_{X,Y}$ and $V_{P,Q}$ is given by

$$V_X = \frac{g_{mCasc}Z_T}{(1 + g_{mfloat}Z_T)} V_P \quad \text{and} \quad V_Y = \frac{g_{mCasc}Z_T}{(1 + g_{mfloat}Z_T)} V_Q. \quad (4)$$

Assuming, $g_{mfloat}Z_T > 1$, the gain of the cascode-driven mesh, $A_{V,mesh}$, deduced from (4) is given by

$$A_{V,mesh} = \frac{V_X}{V_P} = \frac{V_Y}{V_Q} \approx \frac{g_{mCasc}}{g_{mfloat}}. \quad (5)$$

This mesh gain is essentially a scale factor that lowers the swing on $V_{P,Q}$ compared with $V_{X,Y}$. This makes it easier for G_{Mb} to support it without sacrificing its loop gain. Furthermore, similar to (2), the input impedance of the cascode-driven mesh is given by

$$Z_{in,casc} = \frac{V_P}{I_{b1}} = \frac{V_Q}{I_{b2}} = \frac{1 + g_{mfloat}Z_T}{g_{mCasc}} \approx \frac{Z_T}{A_{V,mesh}}.$$

The $A_{V,mesh}$ lowers the input impedance of (2) by the same amount it lowers the swing $V_{X,Y}$. However, this loading effect can be eliminated either by increasing Z_T using cascoding or gain boosting, or by increasing bias current of G_{Mb} which also eases frequency compensation. Similar to (3), the open-loop voltage gain with a cascode-driven mesh is determined using (5) and (6) as

$$G_{casc} = \frac{V_{out}}{V_{in}} = G_{Mb} \left(\frac{Z_T}{A_{V,mesh}} \right) \left(\frac{g_{mCasc}}{g_{mfloat}} \right) (G_{Mo}R_L) = (G_{Mb}Z_T)(G_{Mo}R_L). \quad (6)$$

Comparing (3) and (6), the gains for both conventional- and cascode-driven mesh are exactly the same.

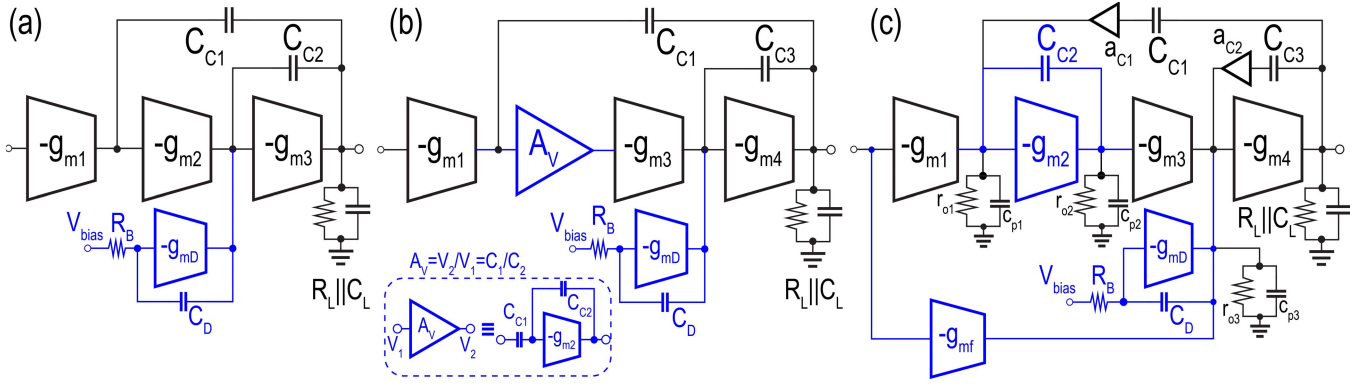


Fig. 7. Topology of (a) constant- Q DFC-NMC [11], (b) adding a gain-stage A_V to increase the loop gain, and (c) proposed damping factor controlled MHNMC.

D. Impact on Nonlinear Distortion

The cascodes M_3 and M_4 relax the swing requirement of G_{Mb} and also make the swing symmetric around output common mode. Fig. 6 shows $A\beta$ at 20 Hz and HD3 as a function of output swing. Here, a conventional MHNMC stabilized amplifier with a conventional class-AB output stage [12] is compared with the proposed amplifier. Both amplifiers are simulated with almost similar quiescent power. At dc, the $A\beta$ of the proposed amplifier is higher as the output common mode of G_{Mb} is around the ground. This gives enough overdrive headroom to ensure transistors operate in strong inversion. On the other hand, the output common mode of G_{Ma} is close to HV_{SS} or HV_{DD} as it directly drives the conventional output stage, which is biased for low-power operation. This forces the G_{Ma} to be biased in moderate or even weak inversion causing reduction in the loop gain in Fig. 6(a) at dc. Besides, transistors in strong inversion have lower distortion coefficient [14]. Thus, the proposed amplifier has better HD3 even at the low-output swing. Furthermore, with increasing output voltage swing, the swing in $V_{X,Y}$ also increases. All the transistors of G_{Ma} that drive V_X and V_Y are exposed to this large and asymmetric swing forcing them to switch their region of operation. This creates significant nonlinear distortion reflected by a rapid increase of HD3 in Fig. 6(b). Contrarily, the relaxed and symmetric output swing of G_{Mb} results in almost flat $A\beta$, which is 20 dB better at $1.8-V_{pp}$ output swing. This leads to 35-dB improvement in HD3. A notch in Fig. 6(b) is created due to the negative term in (1) that cancels HD3 [14]. This term is generated by the second-order harmonic interaction present in a multistage amplifier.

The loop gain and, thus, the distortion improvement from cascode-driven mesh are quite evident in the lower part of the audio band (around 20 Hz). However, it is possible that at higher frequencies (~ 20 kHz), the distortion improvement is limited by the amplifier's frequency response as the loop gain drops rapidly with frequency [15]. Therefore, the frequency compensation that stabilizes the amplifier while achieving higher UGB is equally essential.

IV. FREQUENCY COMPENSATION: DFC-MHNMC SCHEME

The primary requirement of frequency compensation is to keep $A\beta(f)$ high over the audio band while stabilizing the

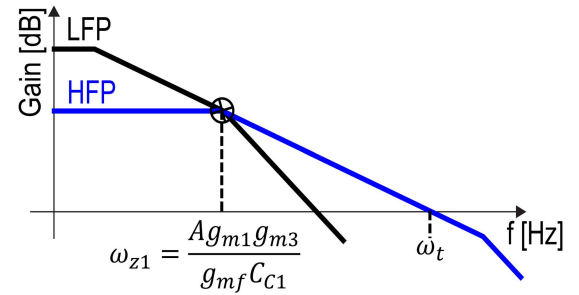


Fig. 8. Frequency response showing LFP and HFP.

amplifier with variation in the load capacitor. The use of DFC is known to improve bandwidth and power efficiency over nested Miller compensation (NMC) [21]. However, it is vulnerable to frequency response peaking and even instability if the load capacitor reduces too low. The constant- Q damping controlled scheme shown in Fig. 7(a) address this issue [11]. It is comprised of g_{mD} , C_D , and R_B . This scheme is highly desirable as it can drive a wide range of load capacitance and handle large voltage swings while achieving fairly high UGB. However, being a three-stage topology, it has limited loop gain. To increase it, a voltage amplifier A_V , as shown in Fig. 7(b), is added. The easiest means to realize such an amplifier is to use a Miller capacitor, C_{C2} , across g_{m2} . This topology is similar to a conventional hybrid NMC (HNMC) [12] with added damping network. The voltage gain C_{C1}/C_{C2} is flat with frequency as long as the non-dominant pole of A_V formed by the input impedance of the integrator $g_{m2}-C_{C2}$ and the Miller capacitor C_{C1} lies above the UGB, ω_t . This is simply written as $g_{m2}/C_{C1} > \omega_t$. Satisfying this condition requires bias current being spent to increase g_{m2} , which increases power consumption. To avoid this, a multipath using g_{mf} can be added in parallel to $g_{m1}-g_{m3}$ yielding DFC-MHNMC topology as shown in Fig. 7(c). Stages $g_{m1}-g_{m4}$, in Fig. 7(c), form a low-frequency path (LFP) that provide high-loop gain at low frequencies while the stages g_{mf} and g_{m4} constitute a high-frequency path (HFP), which stabilizes the amplifier at high frequency by effectively reducing it to a two-stage amplifier. Fig. 8 shows the frequency response of these two paths. The overall response has 20-dB/dec roll-off while benefiting from the loop gain of a four-stage amplifier and UGB of a two stage [22].

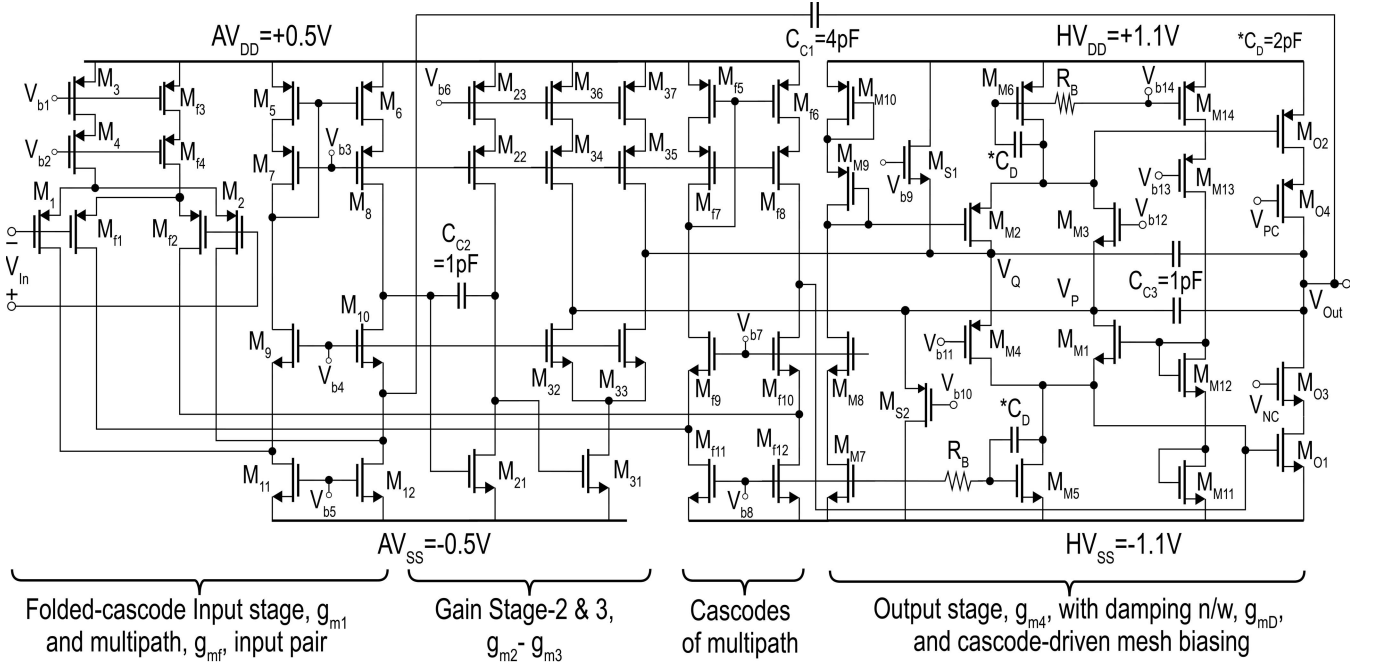


Fig. 9. Transistor-level schematic of the proposed Class-AB amplifier.

Compared to [8], [11], and [17], this topology does not require any loop gain sign inversion to implement Miller compensation as the feedback sign of all the internal loops is implicitly correct. This eliminates unnecessary non-dominant poles and makes the most efficient use of the bias current, as every stage that consumes current also contributes to the loop gain. Furthermore, the cascodes a_{C1} and a_{C2} provide active compensation to cancel the right-half-plane zero of C_{C1} and C_{C2} , respectively, while achieving higher bandwidth than Miller compensation [20]. In Fig. 7(c), the $R_L || C_L$ models the headphone load impedance while the $r_{oi} || c_{oi}$ at the output of $g_{m1}-g_{m3}$ indicate the output impedance and parasitic capacitance. The UGB of DFC-MHNMC ω_t , is constrained by

$$\omega_t = \sqrt{\frac{C_{C1} g_{m3} g_{m4}}{2 C_{C2} c_{p3} C_L}}. \quad (7)$$

The UGB given by (7) is higher by a factor of $[2(C_{C1}/C_{C2})(C_L/c_{p3})]^{1/2}$ compared to the UGB of a conventional MHNMC while consuming the same power [16]. Furthermore, the multipath g_{mf} stage generates a current that only depends upon the input voltage. It acts as an independent signal source that adds a single zero without affecting the poles of the main LFP. Thus, in the presence of multipath, the bandwidth of the LFP can be reduced as it is only responsible for providing high gain. In addition, g_{mf} subdues the impact of the non-dominant pole of A_V on the overall transfer function. Hence, it is no longer required to be placed beyond ω_t . For example, placing near one-tenth of ω_t , i.e., $g_{m2}/C_{C1} > \omega_t/10$, incurs only $\sim 10\%$ reduction in the UGB [22].

As shown in Fig. 8, the multipath adds an independent zero located at

$$\omega_z = \frac{C_{C1} g_{m1} g_{m3}}{C_{C2} g_{mf} C_{C1}}. \quad (8)$$

For a straight first-order frequency response, this zero should cancel the non-dominant pole of the LFP and take over the frequency response. Due to C_{C3} and the damping network, the non-dominant pole also depends upon the load capacitor. Thus, when $C_L = 0$, the takeover criterion is

$$\frac{g_{m1}}{C_{C1}} = \frac{g_{mf}}{C_{C3}} \quad (9)$$

and when $C_L \neq 0$ with $C_L \gg C_{C3}$, the criterion is

$$\frac{g_{m1}}{C_{C1}} = \left(\frac{C_{C3}}{c_{p3}} \right) \frac{g_{m4}}{2 C_L}. \quad (10)$$

While the criterion given by (9) also exists with conventional MHNMC [16], the one that of (10) is specific to DFC-MHNMC due to the presence of a damping network. It is apparent after a simple rearranging of (9) and (10) that the dimensioning of these equations depends on the ratio of transconductance and integrated capacitors. Both of these ratios are well-controlled parameters of a modern CMOS process which, when combined with careful layout techniques, can avoid the formation of pole-zero doublets.

Finally, as the load impedance is only 16Ω and g_{m4} is biased for low power, the gain of the output stage is less than unity ($g_{m4} R_L < 1$). This would reduce the UGB if g_{m4} and C_{C3} were to implement a Miller compensation [23]. By using a_{C2} , active compensation is realized that boosts the UGB by C_{C3}/c_{p3} [20]. This boost in UGB also helps to linearize M_{O1} and M_{O2} thereby improving the distortion performance. As a_{C2} is implemented by reusing the g_m of the cascodes M_3 and M_4 in Fig. 4(a), the bandwidth is improved without an additional power penalty. However, as mentioned in [23], under large signals, the stability of the loop $g_{m4}-C_{C3}-a_{C2}$, shown in Fig. 7(c), degrades as the g_{m4} increases with the signal and grows the loop gain. In [23], additional power consuming circuits are used to address this issue. However, a closer look at

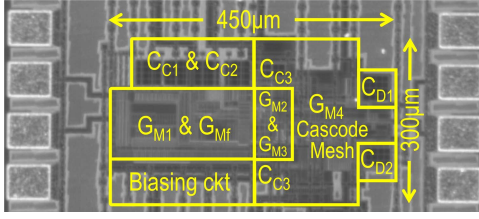


Fig. 10. Bare die micrograph of the test chip.

the implementation of the loop $g_{m4}-C_{C3}-a_{C2}$ reveals that it is naturally stable for an apt choice of g_{mCasc} and I_B . In Fig. 4(a), as the input signal rises so will V_X (V_Y), which increases g_{m4} and steers the bias current I_B to M_4 (M_3). This increases g_m of M_4 (M_3) and stabilizes the active compensation loop.

V. CIRCUIT IMPLEMENTATION

Fig. 9 shows the transistor-level implementation of the topology depicted in Fig. 7(c). To save power, input and the gain stages are powered by ± 0.5 V while only the output stage is powered by ± 1.1 V. Due to the symmetric dual power supply, the output common mode is around 0 V. This permits the amplifier to drive a ground referred load without requiring a large dc-blocking capacitor.

A. Input Stage and Multipath Input Pair

The devices M_1-M_{12} implement a folded cascode input stage, g_{m1} , which is designed for high dc gain and wide UGB provide most of the loop gain. The input pair M_1 and M_2 , and all the current sources, particularly M_{11} and M_{12} are sized to minimize input offset and $1/f$ noise. The multipath input pair, M_{f1} and M_{f2} , are mainly sized to minimize input capacitance. They conduct signal only at high frequency. To improve matching, in addition to the common-centroid layout, the dummy PMOS devices are used around the input and multipath input pairs to overcome well-proximity effects. Monte-carlo simulation estimates the input offset of 200 μ V. A cascoded tail current source M_3 and M_4 ensures that the bias current of the input pairs is held constant while their source node swings with the input signal. This improves the distortion performance. The cascodes M_9 and M_{10} and the Miller capacitor C_{C1} realize active compensation.

B. Gain Stages

The common-source amplifiers, M_{21} and M_{31} realize gain stages g_{m2} and g_{m3} , respectively. While g_{m2} is a high-gain stage, g_{m3} is designed to have a moderate gain but larger output swing. As with the tail current sources of g_{m1} , the cascode current sources ensure a fixed bias current and minimize distortion. A symmetric and balanced drive for the output stage, g_{m4} , suppresses even order harmonics. The cascodes M_{32} and M_{33} generate exactly in-phase, symmetric, and balanced drive signals V_P and V_Q by splitting the signal current of M_{31} . Furthermore, all the compensation capacitors in Fig. 9 are implemented using high-density rotated metal-oxide-metal (RT-MoM) capacitors. To improve matching between the

Miller capacitor C_{C1} and reverse nesting capacitor C_{C2} , the unit cell that constitutes these capacitors are laid out in an array with interdigitated placement in each row and column. The contact terminals are drawn while maintaining the symmetry of the device environment.

C. Class-AB Output Stage

The output stage in Fig. 9 is comprised of the cascode-driven mesh, the damping network, and the output devices. The mesh devices M_{M1} and M_{M2} are biased by a stack of diodes, $M_{M9}-M_{M12}$. The cascodes M_{M3} and M_{M4} , and minimum selectors M_{S1} and M_{S2} are biased by simple cascode-bias generators with bias current supplied from off-chip. All the NMOS devices connected to HV_{SS} are laid out in a separate triple well to avoid interfering with regular devices in the substrate. The output devices M_{O1} and M_{O2} drive the load in a push-pull manner. These devices are sized for equal drive strength in order to cancel the even order harmonic distortion. To protect these devices from gate-drain oxide breakdown, cascodes M_{O3} and M_{O4} are used. The cascode M_{O3} (M_{O4}) and the output device M_{O1} (M_{O2}) share the output voltage as it gets closer to the rail HV_{DD} (HV_{SS}). The cascodes are biased to ensure voltage overstress of <1.1 V on the output devices as the output swings rail-to-rail. Similar to [11], the damping network g_{MD} , is realized by reusing g_m of the tail current sources M_{M5} and M_{M6} along with C_D and R_B . This reuse reduces power consumption while enabling the damping network to operate under large swing present at the gate of M_{O1} and M_{O2} .

D. Cascodes of Multipath Stage

The HFP in Fig. 7(c) comprises of a multipath stage (g_{mf}) directly driving the output NMOS M_{O1} . The g_{mf} has two parts: the input pair M_{f1} and M_{f2} , and the cascode stack $M_{f5}-M_{f12}$. The PMOS active loads $M_{f5}-M_{f8}$ are connected to AV_{DD} while the NMOS current sources $M_{f9}-M_{f12}$ are connected to HV_{SS} . The signal current from the input pair folds into the drain of NMOS current sources M_{f11} and M_{f12} . This arrangement enables voltage domain crossing from ± 0.5 to ± 1.1 V without the need of any level shifters avoiding additional distortion and quiescent power. The AV_{DD} and HV_{SS} voltage difference of 1.6 V is shared by a stack of four devices avoiding over voltage stress.

VI. REALIZATION AND MEASUREMENT RESULTS

The amplifier prototype is fabricated in TSMC 1P9M 65-nm general purpose CMOS and packaged in a commercial plastic QFN 32-leads 5 mm $\times$ 5 mm package. The bare die micrograph is shown in Fig. 10. Multiple bondpads are allocated to $AV_{DD}-AV_{SS}$ and $HV_{DD}-HV_{SS}$ to reduce the IR drop. The output stage is placed close to the $HV_{DD}-HV_{SS}$ and the output pad. The total area of the amplifier is 0.16 mm², of which 50% is occupied by amplifier core while the compensation capacitors occupy only 25%. The remaining area is occupied by the bias generation circuit and the de-coupling capacitors. The test chip is measured using an SRS DS360 signal source that generates a high-purity sinusoidal tone with THD+N less

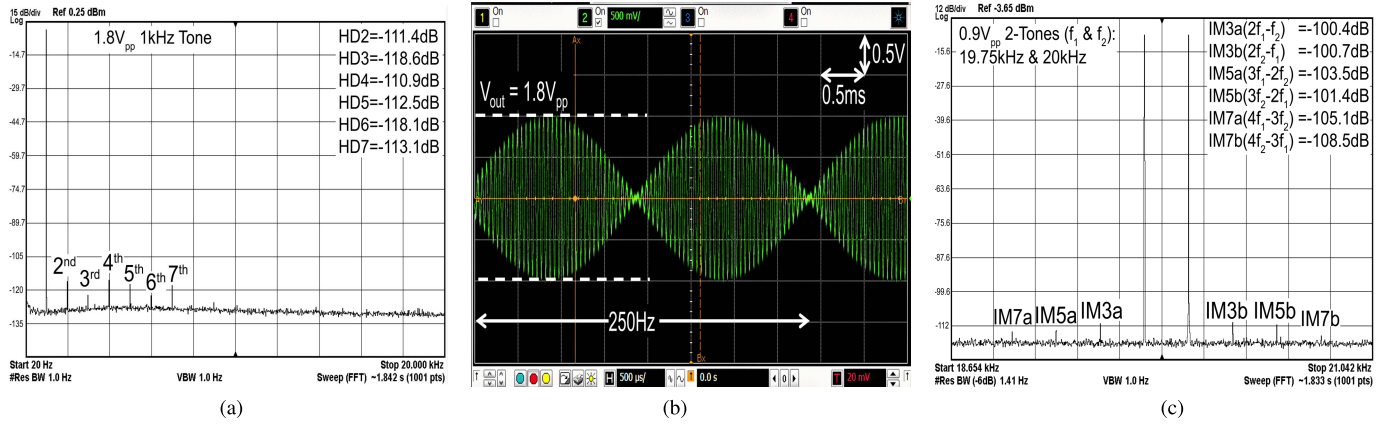


Fig. 11. Measured output spectrum and a 2-tone output transient. The 2-tones are located at 19.75 and 20 kHz. (a) Spectrum of 1-kHz input tone. (b) Transient output for 2-tones input. (c) Spectrum for 2-tone input.

than -110 dB, and a spectrum analyzer that is configured for a dynamic range of more than 120 dB. The test chip is arranged in inverting negative feedback, as shown in Fig. 2(d), with $R_I = 1$ k Ω , $R_2 = 4.7$ k Ω , $R_L = 16\Omega$, $C_F = 12$ pF, and $C_L = 0.33$ nF.

Including core and bias circuits, the amplifier draws 0.31 mA from ± 1.1 V and 0.28 mA from ± 0.5 V while driving a ground-referred 16- Ω ||0.33-nF load. For most of the tests, a closed-loop gain of 13.5 dB and output swing of 1.8 V_{pp} are ensured. Fig. 11 shows the linearity of the amplifier measured with 1-tone (1 kHz) and 2-tone inputs. In Fig. 11(a), the first seven harmonic distortion tones are visible and are all below -110 -dB level. Fig. 11(b) shows the output of the amplifier for a 2-tone input. The amplifier has its lowest loop gain at the high end of the audio band. Hence, the two tones are chosen at 19.75 and 20 kHz. The difference between these two tones generates a beat frequency of 250 Hz in Fig. 11(b). The spectrum for this 2-tone output is shown in Fig. 11(c). The intermodulation (IM) products are <-100 dB level with IM3 tones being the most noticeable. These THD and IM measurements are sensitive to the spectrum analyzer configuration. Hence, their accuracy is cross verified by selectively amplifying distortion from the virtual ground node [24].

Fig. 12(a) shows the THD+N and IM3 with frequency. For frequencies below 20 Hz, the loop gain of 110 dB [Fig. 13(a)] pushes the distortion tones below the noise floor. As a result, THD+N is limited by the flicker noise. With increasing frequency, the flicker noise reduces which improves THD+N. Beyond the 3-dB bandwidth of the amplifier (~ 40 Hz), the loop gain decreases causing THD+N to increase and reach the worst case lowest value. At higher frequencies, the loop gain further reduces, but THD+N appears to improve. This occurs as the distortion tones fall outside of the audio band and are filtered [25]. As shown in Fig. 12(a), the distortion performance indeed is worsened by the loss of loop gain. The 2-tones placed closely near 20 kHz create the IM cross-products within the audio band. The loss of loop gain at higher frequencies makes the IM3 increase from -111 to -100 dBc. Fig. 12(b) shows the THD+N measured versus input amplitude of a 1 kHz tone. For small input amplitudes, the distortion tones

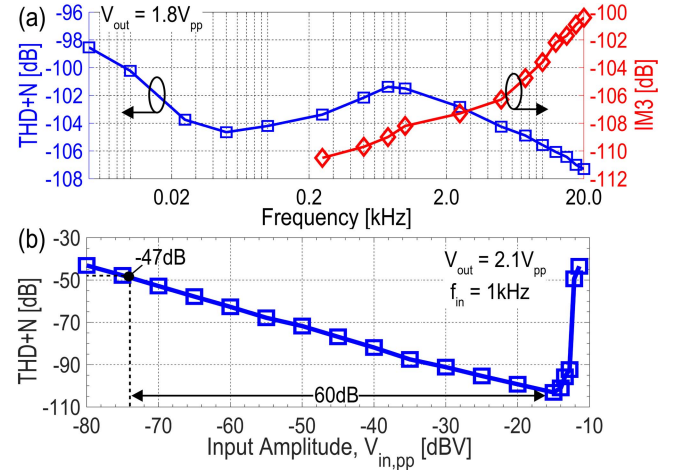


Fig. 12. Measurement (a) THD+N and IM3 with frequency and (b) THD+N with input amplitude. During both measurements, a 16 Ω ||0.33-nF load is used.

are below the noise floor. Hence, as argued earlier, THD+N is limited by the noise, creating a decreasing trend. The measured dynamic range at a -60 -dB input signal level is 107 dB. The plot also shows that the 1% THD power level occurs at the input amplitude of 0.45 V_{pp} ($V_{Out,pp} = 2.1$ V). Beyond this point, the output clips and creates excessive hard distortion.

The ac response of the amplifier is shown in Fig. 13(a). The $A\beta$ until 20 kHz is higher than 57 dB, which is required to minimize distortion in the audio band. The phase margin (PM) is 55° and the gain margin (GM) is 11 dB, while the UGB is measured to be 12.3 MHz. The transient step response shown in Fig. 13(b) and (c) along with higher UGB achieved confirms the effectiveness of the proposed frequency compensation scheme in stabilizing the amplifier with a capacitive load as large as 10 nF. No ringing is observed in either a rising or falling step as C_L is varied from no load to 10 nF. At 22 nF, the stability of the amplifier is degraded, which results in ringing. Fig. 13(d) and (e) shows the amplifier response to a large signal input. While the output swings to $2.2V_{pp}$, the rising edge and falling edge slew

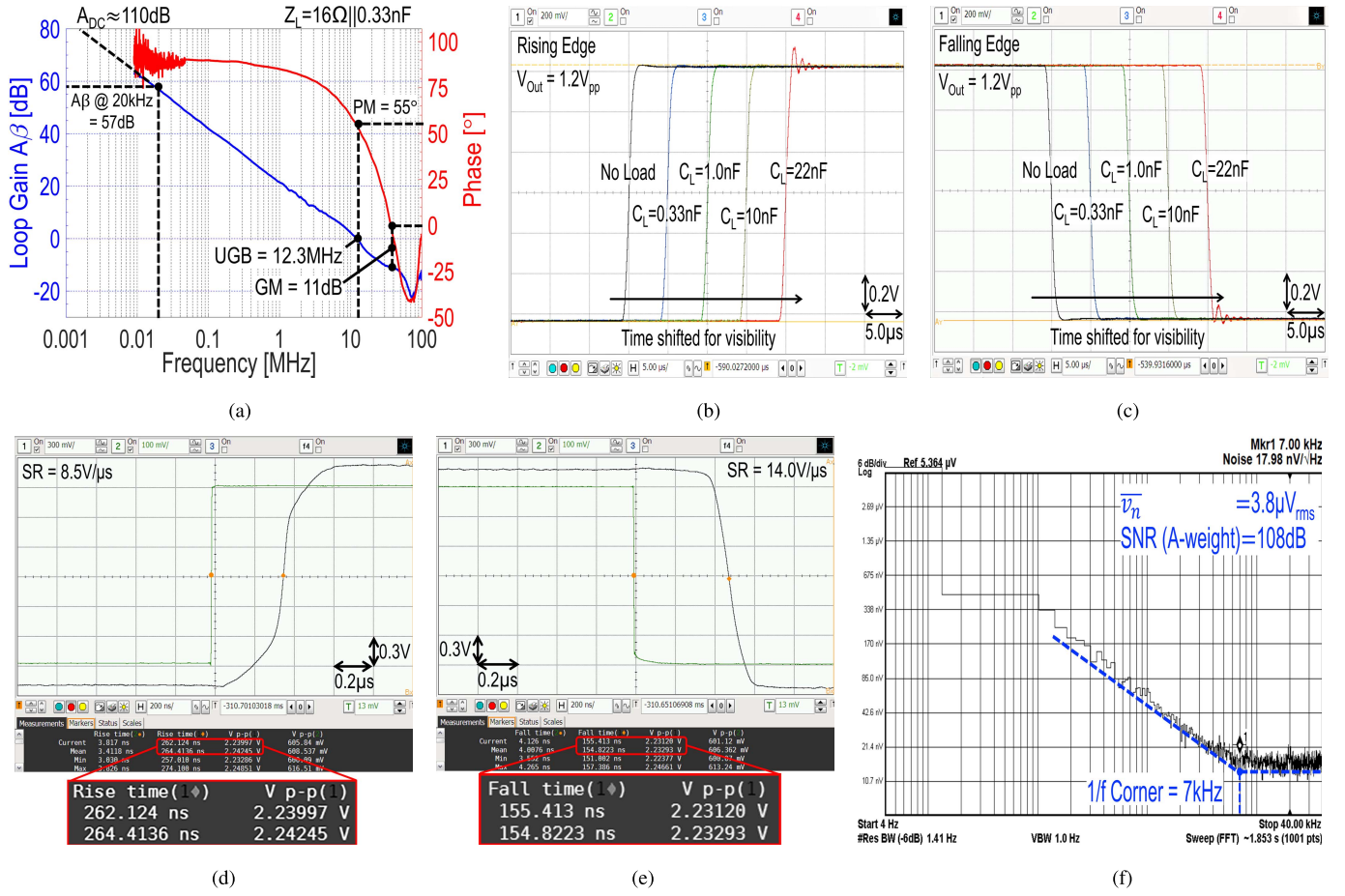


Fig. 13. Measured ac-response, transient response with varying C_L from no load to 22 nF , slew rate for a fast large input signal and noise PSD. (a) Loop gain ($A\beta$) and phase of the amplifier. (b) Rising edge step response. (c) Falling edge step response. (d) Rising edge slew rate with $C_L = 0.33\text{ nF}$. (e) Falling edge slew rate with $C_L = 0.33\text{ nF}$. (f) Measured noise spectrum and A-weighted SNR.

TABLE I

PERFORMANCE SUMMARY AND COMPARISON WITH STATE-OF-THE-ART MODERATE TO HIGH LINEARITY CLASS-AB AMPLIFIERS

Reference	This Work	[8] JSSC'17	[11] JSSC'09	[3] JSSC'14	[2] JSSC'12	[17] VLSI'12	[6] ADI'17
Process	65nm CMOS	65nm CMOS	130nm CMOS	40nm CMOS	130nm CMOS	40nm CMOS	XFCB Bipolar
Supply [V]	1.0/2.2	2.0	1.2/2.0	3.1-4.5	1.5	1.8	6.6
Output Swing [V_{pp}]	1.8	1.5	1.6	3.24	2.5	1.4	2.8
Quiescent Power [mW]	0.97	0.4	1.2	3.8	1.5	1.8	19.2
Peak Load Power [mW]	51.2	35	40	164	60	31.2	45
Load ($R_L C_L$)	$16\Omega (0\text{-}10\text{nF})$	$16\Omega (0\text{-}2\text{nF})$	$16\Omega (1\text{pF}\text{-}22\text{nF})$	$16\Omega \text{---} 600\Omega^{**}$	$16\Omega \text{---} 32\Omega^{**}$	$16\Omega^{**}$	$16\Omega (0.33\text{-}2.2\text{nF})$
Loop-gain ($A\beta$) @20kHz [dB]	57	39 [‡]	46.1 [‡]	—	—	—	62
UGB [MHz]	12	1.78	4	—	—	—	25
Slew Rate [$\text{V}/\mu\text{s}$]	8.5	1.1	0.4	—	—	—	14
SNR (A-weighted) [dB]	108	93.5	92	100	98	100	110
THD+N@max.	Lowest	-107.3	-93*	—	—	-96	-110.4
Output [dB]	Full-Audio band	-101.4	-89	-84	-72	-91	-106
Linearity FOM ([17] & [8]) [†]	62,015	24,774	5283	6848	2600	6150	4700

** Load cap value not available. * Extracted from plots. [‡] Derived from UGB assuming single dominant pole.

[†] FOM = Peak Load Power/(Quiescent Power · THD+N[%])

rate are 8.5 and $14\text{ V}/\mu\text{s}$, respectively. This yields a worst case slew limited power bandwidth of 1.5 MHz [$f_p = \text{Slew rate}/(\pi V_{Out,pp})$], which is more than sufficient to avoid slew induced distortion over the audio band [26]. The measured

input referred noise, integrated from the noise spectrum shown in Fig. 13(f), is about $3.8\mu\text{V}_{rms}$, which matches closely with the simulation. The resulting SNR of 108 dB is high enough that it does not corrupt the overall THD+N.

Table I summarizes the amplifier performance and compares it with the recent state-of-the-art CMOS and non-CMOS class-AB amplifiers. To emphasize high linearity at low-quiescent power, the figure of merit (FOM) suggested in [8] and [17] is adopted. Among the reported amplifiers designed in the standard CMOS process, the proposed amplifier achieves the highest linearity and dynamic range. It consumes sub-mW quiescent power while delivering 51.2-mW peak load power, which compares fairly with other designs. Thanks to the proposed frequency compensation scheme, the measured UGB of 12.3 MHz is $7\times$ higher than [8] and $3\times$ higher than [11]. This translates to 57 dB gain at 20 kHz, which improves THD+N by 13 dB [8] and 18 dB [11]. Also, the measured SNR of 108 dB is >15 dB higher than the nearest state-of-the-art designs. The improved linearity results in the highest reported linearity FOM among Class-AB drivers. Compared to a recent hi-fi amplifier product [6], the proposed amplifier achieves similar linearity and performance but at one-twentieth power without using any specialized fabrication process. When measured over eight chips from the same wafer with fix load of $16\ \Omega || 0.33\ \text{nF}$, the mean peak THD+N is -102.5 dB with $<3\%$ spread, the mean PM is 57.9° with $<7\%$ spread, and the mean GM is 13.8 dB. All the eight chips are stable at no load.

VII. CONCLUSION

A class-AB amplifier is described that achieves <-100 dB THD+N at <1 mW. Two key techniques are used to enable this performance. A cascode-driven mesh biasing technique is shown to limit the number of high-swing nodes in the output stage to exactly two. It lowers the output swing requirement of the gain stages and thus avoids loop-gain degradation. The DFC-MHNMCM scheme stabilizes the amplifier while achieving 12.3-MHz UGB. As a result, the loop gain at 20 kHz is increased, which further suppresses distortion. In addition, several circuit techniques and design choices are made to avoid distortion and power from extra circuits such as level shifters, sign inversion, and so on. Combining these approaches, the proposed amplifier achieves the highest THD+N ever reported among CMOS sub-mW power consuming class-AB amplifiers. Replacing the mean THD+N of -102.5 dB for that of the integrated amplifier in Fig. 1, the audio path can now achieve a linearity of 100 dB THD+N with a reduced power of 1.7 mW. This meets the hi-fi audio requirements and can potentially enable a hi-fi audio path in the future mobile devices.

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